Assignment 3 - Demon Slayer Corps: The String Hunt

Roll No: 24i-3126

Final Selection - test_matchers.py (24/24 PASS)

```
zain-shykh@zain-shykh-IdeaPad-1-15IAU7:~/Desktop/Algo_Assignements/DemonHunt_data/starter$ python3 test_matchers.py
  PASS naive       aba in ababa
  PASS rabin-karp  aba in ababa
  PASS kmp         aba in ababa
  PASS automaton   aba in ababa
  PASS naive       ababaca in abababacaba
  PASS rabin-karp  ababaca in abababacaba
  PASS kmp         ababaca in abababacaba
  PASS automaton   ababaca in abababacaba
  PASS naive       aa in aaaaaa
  PASS rabin-karp  aa in aaaaaa
  PASS kmp         aa in aaaaaa
  PASS automaton   aa in aaaaaa
  PASS naive       abcab in acaabcaabcab
  PASS rabin-karp  abcab in acaabcaabcab
  PASS kmp         abcab in acaabcaabcab
  PASS automaton   abcab in acaabcaabcab
  PASS naive       d in abcabcabc
  PASS rabin-karp  d in abcabcabc
  PASS kmp         d in abcabcabc
  PASS automaton   d in abcabcabc
  PASS prefix function of ababaca
  PASS rabin-karp shifts with d=10, q=11
  PASS rabin-karp spurious hits with d=10, q=11
  PASS kmp resume across a chunk boundary
```

Mission 1 - The Crow's Alarm (KMP)

```
zain-shykh@zain-shykh-IdeaPad-1-15IAU7:~/Desktop/Algo_Assignements/DemonHunt_data/starter$ python3 mission1.py
  pi = [0, 0, 1, 2, 3, 4]
  Alarm count: 415
  First 10 shifts: [8620, 8622, 8624, 39118, 39126, 88124, 88132, 90213, 90215, 90217]
zain-shykh@zain-shykh-IdeaPad-1-15IAU7:~/Desktop/Algo_Assignements/DemonHunt_data/starter$
```

Mission 2 - Tamayo's Blood Lab (finite automaton)

```
zain-shykh@zain-shykh-IdeaPad-1-15IAU7:~/Desktop/Algo_Assignements/DemonHunt_data/starter$ python3 mission2.py
  delta for GATTACAGATTC (rows = states 0..12)
  state |   A   C   G   T
      0 |   0   0   1   0
      1 |   2   0   1   0
      2 |   0   0   1   3
      3 |   0   0   1   4
      4 |   5   0   1   0
      5 |   0   6   1   0
      6 |   7   0   1   0
      7 |   0   0   8   0
      8 |   9   0   1   0
      9 |   0   0   1  10
     10 |   0   0   1  11
     11 |   5  12   1   0
     12 |   0   0   1   0

  Reverse complement: GAATCTGTAATC

  sample_01.txt: forward=0 reverse=0
  sample_02.txt: forward=0 reverse=0
  sample_03.txt: forward=0 reverse=0
  sample_04.txt: forward=0 reverse=0
  sample_05.txt: forward=0 reverse=0
  sample_06.txt: forward=0 reverse=0
  sample_07.txt: forward=0 reverse=3
  sample_08.txt: forward=0 reverse=0
  sample_09.txt: forward=0 reverse=0
  sample_10.txt: forward=0 reverse=0

  Demon in disguise: sample_07.txt
zain-shykh@zain-shykh-IdeaPad-1-15IAU7:~/Desktop/Algo_Assignements/DemonHunt_data/starter$
```

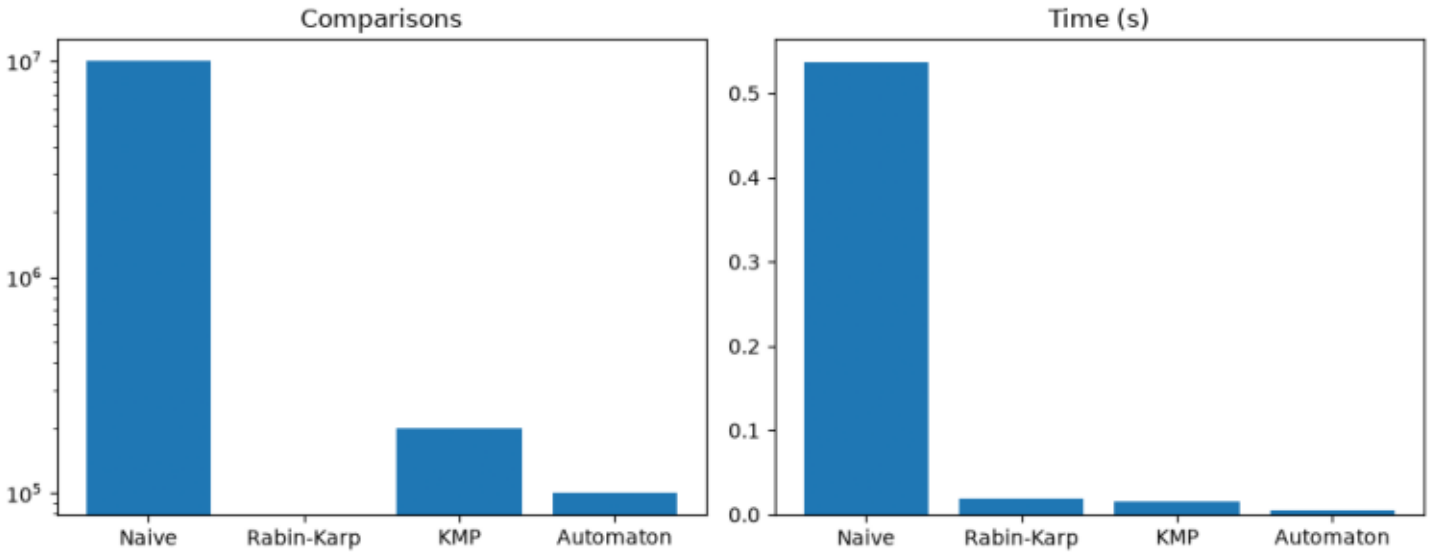
Mission 3 - The Copycat Slayer (Rabin-Karp)

```
zain-shykh@zain-shykh-IdeaPad-1-15IAU7:~/Desktop/Algo_Assignements/DemonHunt_data/starter$ python3 mission3.py
  Similarity ranking (highest first):
  1. Zenitsu   89.80%
  2. Kanao     31.86%
  3. Inosuke   14.96%
  4. Tanjiro    0.80%
  5. Genya      0.80%
zain-shykh@zain-shykh-IdeaPad-1-15IAU7:~/Desktop/Algo_Assignements/DemonHunt_data/starter$
```

Mission 3: why the word-changer scores lower than deserved

A 40-character window counts only if it matches the manual exactly, so one changed word breaks every window covering it. Rabin-Karp finds exact copies only, not paraphrased ones.

Bonus - Breathing Styles Tournament



Winner and loser

Winner: the automaton does exactly one lookup per text character (100,000), with no backtracking.
KMP is close: it never moves back in T, so it needs at most 2n comparisons.
Loser: naive restarts at every shift and matches 99 a's before failing, so about n*m comparisons.